Quantifying Moderator Effects on High-Dimensional Multi-Subject Time Series

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Abstract

Researchers in many scientific fields use time series data to estimate the causal effects that variables have on each other over time. These relationships can be modeled via Granger-causal networks (GCN), where nodes correspond to variables, and directed links indicate lagged temporal relationships among them. When longitudinal data is gathered over multiple subjects, GCN models must account for potential subject heterogeneity. While methods in the literature can perform estimation of heterogeneous GCNs, they often lack interpretable explanations of the source of such measured heterogeneity. In the present paper, we propose the moderator vector autoregressive model (MOD-VAR) for interpretable estimation of heterogeneous GCN in multi-subject longitudinal data. We achieve this by using external moderator variables to explain the heterogeneity of GCNs across subjects. Our theoretical results confirm the improved estimation of MOD-VAR for heterogeneous GCN estimation in high dimensional scenarios, even when the GCN are highly heterogeneous. Furthermore, our extensive numerical simulations confirm the superior performance of our proposed MOD-VAR for GCN estimation and forecasting in a wide variety of settings. Analyzing a resting state fMRI dataset with time-invariant moderators, we demonstrate that MOD-VAR recovers moderating effects of age and locus coeruleus structural integrity (volume and contrast-to-noise ratio) on signaling within and between the salience, frontoparietal, and default mode networks.

Keywords: Adaptive estimation, network regression, time series, vector auto-regressive model, weighted networks. The moderation effect

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1 Introduction

The modeling and forecasting of high-dimensional multi-subject time series data has been a long-standing area of research in statistics and machine learning [11], with applications in economics, finance, neuroscience, and psychology [23; 14; 25; 5; 15], among other areas. *Granger-causal networks* (GCN) are a particularly useful way to describe the relations among variables in a time series. In a GCN, nodes correspond to variables, and a directed link between two nodes represents an effect of the origin variable in the present to future values of the target variable. When the time dependence is linear, the GCN can be fully recovered by estimating coefficients of a sparse vector auto-regressive model (VAR). This has led to the development of a wide variety of methods aimed at accurately estimating GCNs in time series data [2; 3; 22]. For a review, see Wei [31].

One major statistical challenge is that of estimating GCNs over multiple subjects. For example, in functional magnetic resonance imaging (fMRI) studies, researchers are often interested in determining the patterns of connectivity among different regions of the brain [14; 25; 5]. By processing fMRI data, researchers can measure the activity of brain regions over time. In this context, a GCN may describe the connectivity of electrical signals across different brain regions. In multi-subject studies, we expect the connectivity patterns to have shared components, while also showing some heterogeneity across different subjects. In this setting, fitting a single VAR model may lead to bias and inaccurate forecasting. On the other hand, training separate VAR models for each subject may miss the chance to exploit the common structure to obtain improved forecasting accuracy [6].

Several works have explored the problem of fitting and testing models for multi-subject high dimensional time series. The Multi-VAR method [7; 8] applies penalized estimation to recover both joint and individual components of GCNs across multiple subjects. The group iterative multiple model estimation procedure (GIMME) creates individual networks with unified structural equation models for each subject and then identifies majority-shared network edges [9; 15; 16]. Recent works have also explored the task of testing the significance of sparse GCN estimators. For example, Zhu and Liu [35]; Uematsu and Yamagata [28] developed a multiple testing procedure based on the debiased lasso [29; 32; 13] to determine if there were significant differences in GCNs across multiple subjects.

Despite the progress of prior works, a key question remains unanswered: *Can we explain the heterogeneity that exists across subjects?* For example, in the aforementioned example of fMRI data, researchers may be interested not only in detecting heterogeneity in the structures of the GCNs across individuals, but to determine whether external moderators like age, biological sex, or existing health conditions may explain that heterogeneity. Existing methods that measure GCN heterogeneity are not currently capable of determining if the variation observed in the GCNs can be explained by such external moderators.

Although models that link the GCN of time series with moderator variables have been explored via multilevel modeling [26; 17], such methods have limited applicability since they are unable to handle a large number of variables d or moderator variables p unless the number of subjects or time-measurements per subjects is prohibitively large. Alternatively, the vector autoregressive model with exogenous variables (VARX) incorporates the effect of external moderators on the VAR forecasting model [22; 21]. Despite its advantages, external moderators are only considered as potentially affecting the mean behavior of the time series. Therefore, the VARX does not consider the potential effect that moderators have on the GCNs themselves. Thus, there is a pressing need for methods that can estimate the influence that moderator variables have on the GCNs across multiple subjects, and can provide reliable estimates, even in cases where the number of parameters may be significantly larger than the number of observations.

In the present work, we propose the moderator vector autoregressive model (MOD-VAR) for the simultaneous estimation of GCNs over multi-subject time-series data with time-invariant moderators. Our method measures the effect that external moderators may have on the GCNs across datasets. We achieve this by decomposing each GCN for each individual into (*i*) connections shared across all datasets, (*ii*) connections in the network dependent on external moderators, and (*iii*) connections that are fully individual, and cannot be explained by the given moderator variables. Our method is highly general, encompassing the VAR and Multi-VAR frameworks [7; 8]. Although similar moderator models have been explored in the context of Gaussian graphical models [20; 33; 18], to the best of our knowledge this is the first method to decompose an autoregressive structure into shared, moderated, and individual components for enhancing the forecasting and interpretability of multi-subject high-dimensional time series.

The rest of the paper is organized as follows. In section 2 we introduce our model of moderator time series model, and introduce our MOD-VAR estimator. In section 3 we provide details on the implementation of our proposed method. In Section 6 we provide numerical comparisons of our proposed MOD-VAR with existing methods found in the literature. In Section 7 we present the analysis of a dataset of brain activation across multiple regions via fMRI measurements, and quantify the effect that multiple moderators have on brain connectivity across subjects. Closing remarks and discussions are found in Section 8.

Notational Conventions

We establish the following notational conventions. For any n, let $[n] = \{1, 2, \dots, n\}$. For any $k \leq \ell$, we denote $[k : \ell] = \{k, k+1, \dots, \ell-1, \ell\}$. For any $\mathrm{A} \in \mathbb{R}^{p\times q}$, we denote the transpose of A as A$'$. For $\boldsymbol{x} = (x_1, x_2, \dots, x_d)' \in \mathbb{R}^d$, we denote by $\|\boldsymbol{x}\|_1, \|\boldsymbol{x}\|_2, \|\boldsymbol{x}\|_\infty$ the ℓ_1-norm, euclidean norm and entrywise maximum norm respectively, given by $\|\boldsymbol{x}\|_1 = \sum_{i=1}^d |x_i|$, $\|\boldsymbol{x}\|_2 = \sqrt{\boldsymbol{x}'\boldsymbol{x}}$, and $\|\boldsymbol{x}\|_\infty = \max_{i\in[d]} |x_i|$. For any $\Sigma = (\sigma_{ij})_{i,j=1}^d \in \mathbb{R}^{d\times d}$, we denote the trace as $\mathbf{tr}(\Sigma) = \sum_{i\in[d]} \sigma_{ii}$. For $\mathrm{A} \in \mathbb{R}^{p\times q}$, we denote by $|||\mathrm{A}|||_2$, and $|||\mathrm{A}|||_F$ the ℓ_2 operator, and Frobenius norms, respectively, given by $|||\mathrm{A}|||_2 := \sup_{\boldsymbol{x}\in\mathbb{R}^q\setminus\{\mathbf{0}\}} \|\mathrm{A}\boldsymbol{x}\|_2/\|\boldsymbol{x}\|_2$, and $|||\mathrm{A}|||_F = \sqrt{\mathbf{tr}(\mathrm{A}'\mathrm{A})}$. Additionally, we denote by $\|\mathrm{A}\|_1$ the vectorized ℓ_1-norm of the matrix A, given by $\|\mathrm{A}\|_1 = \sum_{i,j} |A_{ij}|$. For any k, we denote $\mathbb{1}_k \in \mathbb{R}^k$ the vector with all one entries. For any set A, we denote by $\mathbb{1}(A)$ the indicator function of A. For any $x \in \mathbb{R}$, we denote $\mathrm{sign}(x) = \mathbb{1}(x > 0) - \mathbb{1}(x < 0)$. For any symmetric matrix $\Sigma \in \mathbb{R}^{d\times d}$, we denote the minimum and maximum eigenvalues of Σ as $\Lambda_{\min}(\Sigma), \Lambda_{\max}(\Sigma) \in \mathbb{R}$, respectively.

2 Moderator Vector Autoregressive Model

2.1 Vector Autoregressive Models

Consider a study with n subjects. For each subject $k \in [n]$, we have access to the measurements of d different variables of interest over T_k equally spaced time points, which we denote by $\mathbf{Y}_k = \{\boldsymbol{Y}_t^{(k)}\}_{t=1}^{T_k} \subset \mathbb{R}^d$. We model the lagged temporal relationships that exist between the different variables, by a VAR(1) model, expressed as

$$\boldsymbol{Y}_t^{(k)} = \Phi^{(k)*}\boldsymbol{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 2 \leq t \leq T_k,$$

where $\Phi^{(k)*}$ is a $d \times d$ matrix that represents the true lag-1 multivariate relationships for the subject k. In particular, for $1 \leq i, j \leq d$, the entry $\Phi_{ij}^{(k)*}$ represents the effect that the variable j at time $(t-1)$ affects the variable i at time t. Notice that we define a distinct coefficient matrix $\Phi^{(k)*}$ for each subject $k \in [n]$. Given the matrix $\Phi^{(k)*}$, we define the GCN associated with subject k, as the network with set of nodes $N = \{1, 2, ..., d\}$, and a link $i \to j$ is present in the GCN if and only if $\Phi_{ji}^{(k)*} \neq 0$. Thus, the structure of the GCN depends on the sparsity pattern of the matrix $\Phi^{(k)*}$.

For each subject $k = 1, 2, \ldots, n$, in addition to the time series data $\mathbf{Y}_k$, we have a set of p potential moderator variables $\boldsymbol{X}_k = (X_{k1}, X_{k2}, \ldots, X_{kp})' \in \mathbb{R}^p$. Notice that the moderators in $\boldsymbol{X}_k$ are not time-varying, and thus can be thought as stable features of each subject. For example, in a neuroimaging study analyzing brain connectivity, these moderators may include clinical or demographic variables such as biological sex or age. We may also consider controlled experimental variables of interest such as treatment status of individual, or magnitude of the dose provided to a patient. Our goal in this work is to determine if such moderator variables explain the heterogeneity in the GCNs across subjects, and exploit this potential effects to improve the estimation of the GCNs.

2.2 Moderator Vector Autoregressive Models

We first introduce the scenario in which all cross-subject heterogeneity can be explained by the moderators in $\boldsymbol{X}_k$. In this case, we can represent the moderator-dependent relationships as $\Phi^{(k)*} = \Phi^*(\boldsymbol{X}_k)$, where, for any vector of p moderators $\boldsymbol{x} = (x_1, x_2, \ldots, x_p)'$,

$$\Phi^*(\boldsymbol{x}) := \Phi^{(0)*} + x_1\Phi^{(M1)*} + x_2\Phi^{(M1)*} + \ldots + x_p\Phi^{(Mp)*}, \tag{1}$$

where $\Phi^{(0)*}, \Phi^{(M1)*}, \Phi^{(M2)*}, \ldots, \Phi^{(Mp)*} \in \mathbb{R}^{d\times d}$. The matrix $\Phi^{(0)*} \in \mathbb{R}^{d\times d}$ represents the shared connectivity in the GCN across subjects. For each moderator variable $1 \leq \ell \leq p$, the matrix $\Phi^{(M\ell)*}$ represents the effect that the ℓ-th moderator has on the time series dynamic. More specifically, for $i, j \in [d]$ and $\ell \in [p]$ the entry $\Phi_{ij}^{(M\ell)*}$ represents the effect that the moderator variable ℓ has on the lagged relationship of j on the variable i. If for a subject i we have $x_{i\ell} = 0$, then $x_{i\ell}\Phi^{(M\ell)*} = \mathbf{0}$ and thus, the moderator ℓ does not have any effect on the time dynamics for subject i. In addition, the large values of $x_{i\ell}$ are associated with stronger effects of the moderator ℓ on the dynamic of subject i. Under the model (1), we define the moderator vector auto-regressive (MOD-VAR) model via the equation

$$\boldsymbol{Y}_t^{(k)} = \Phi^*(\boldsymbol{X}_k)\boldsymbol{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 1 \leq k \leq n \text{ and } 2 \leq t \leq T_k. \tag{2}$$

We illustrate our MOD-VAR model from (2) with the following example.

Example 1 (Binary moderators)**.** *We consider a VAR(1) model with n subjects. For each subject, we measure a single moderator variable (i.e., $p = 1$) $X_1, X_2, \ldots, X_n \in \mathbb{R}$ with $\{X_k\}_k \overset{i.i.d.}{\sim}$ Bernoulli(p). In this case, our propose model can be written as,*

$$\boldsymbol{Y}_t^{(k)} = [\Phi^{(0)*} + X_{k1}\Phi^{(M1)*}] \cdot \boldsymbol{Y}_{t-1}^{(k)} + \boldsymbol{e}_t^{(k)}$$

Notice that the GCN of a subject $k \in [n]$ incorporates the matrix $\Phi^{(0)*} \in \mathbb{R}^{d\times d}$ shared across all subjects. Furthermore, there is an individual structure that depends on the value of X_k. All subjects with $X_k = 1$ will have GCN $\Phi^{(k)*} = \Phi^{(0)*} + \Phi^{(M1)*}$, while subjects with $X_k = 0$ have GCN $\Phi^{(k)*} = \Phi^{(0)*}$. Thus, the matrix $\Phi^{(M1)*}$ represents the structural differences in the GCN across the the grouping given by the moderators $X_1, X_2, \ldots, X_n$.

Now, consider modeling two binary moderator variables $\boldsymbol{X}_k = (\boldsymbol{X}_{k1}, \boldsymbol{X}_{k2})^\top \in \{0,1\}^2$, we have now a total of four potential GCNs: (i) $\Phi^{(k)*} = \Phi^{(0)*}$ if $\boldsymbol{X}_k = (0,0)^\top$, (ii) $\Phi(k)* = \Phi^{(0)*} + \Phi^{(M1)*}$ if $\boldsymbol{X}_k = (1,0)^\top$, (iii) $\Phi(k)* = \Phi^{(0)*} + \Phi^{(M2)*}$ if $\boldsymbol{X}_k = (0,1)^\top$, and (iv) $\Phi(k)* = \Phi^{(0)*} + \Phi^{(M1)*} + \Phi^{(M2)*}$ if $\boldsymbol{X}_k = (1,1)^\top$. We illustrate the GCNs of two subjects with different moderator values in Figure 1.

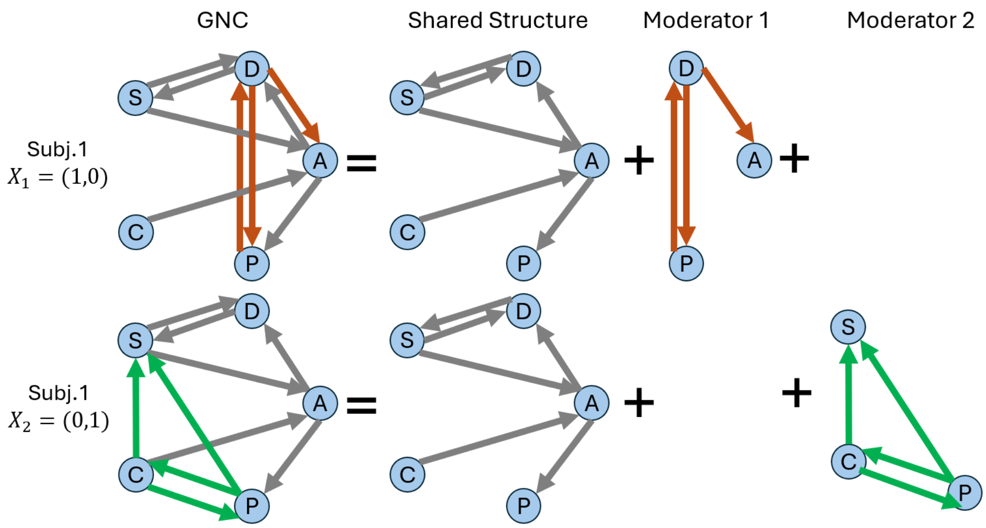


Figure 1: Illustration of the MOD-VAR framework with $p = 2$ binary moderator variables, as discussed in Example 1. Top-panels: visualization of the decomposition of the GCN of subject 1. Since $\boldsymbol{X}_1 = (1, 0)^\top$, only the first moderator has an effect on the GCN for this subject. Bottom panels: visualization of the GCN for subject 2. Since $\boldsymbol{X}_2 = (0, 1)^\top$, only the second moderator affects the GCN structure.

2.3 MOD-VAR with Fully Individual Connections

While external moderators may account for some of the GCN heterogeneity across subjects, often some heterogeneity may remain unexplained. Thus, in addition to a shared component $\Phi^{(0)*}$, and moderator effects $\Phi^{(M1)*}, \dots, \Phi^{(Mp)*}$, we consider $\boldsymbol{\Delta}^{(I1)*}, \boldsymbol{\Delta}^{(I2)*}, \dots, \boldsymbol{\Delta}^{(In)*} \in \mathbb{R}^{d\times d}$ matrices corresponding to fully individual GCN connections. For each subject k, the coefficient matrix $\Phi^{(k)*}$ can be decomposed as

$$\Phi^{(k)*} = \Phi^*(\boldsymbol{X}_k) + \boldsymbol{\Delta}^{(Ik)*} = \Phi^{(0)*} + \left(\sum_{\ell=1}^{p} X_{k\ell}\Phi^{(M\ell)*}\right) + \boldsymbol{\Delta}^{(Ik)*}. \tag{3}$$

Notice that $\boldsymbol{\Delta}^{(Ik)*} \in \mathbb{R}^{p\times p}$ only appears on the GCN of subject k. Thus, it represents the individual connections for subject k not explained by moderator data. We illustrate the role of the additional fully individual structure in the GCNs across subjects in Figure 2.

To simplify using the notation introduced in (1), we write,

$$\boldsymbol{Y}_t^{(k)} = (\Phi^*(\boldsymbol{X}_k) + \boldsymbol{\Delta}^{(Ik)*})\boldsymbol{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 1 \le k \le n \text{ and } 2 \le t \le T_k. \tag{4}$$

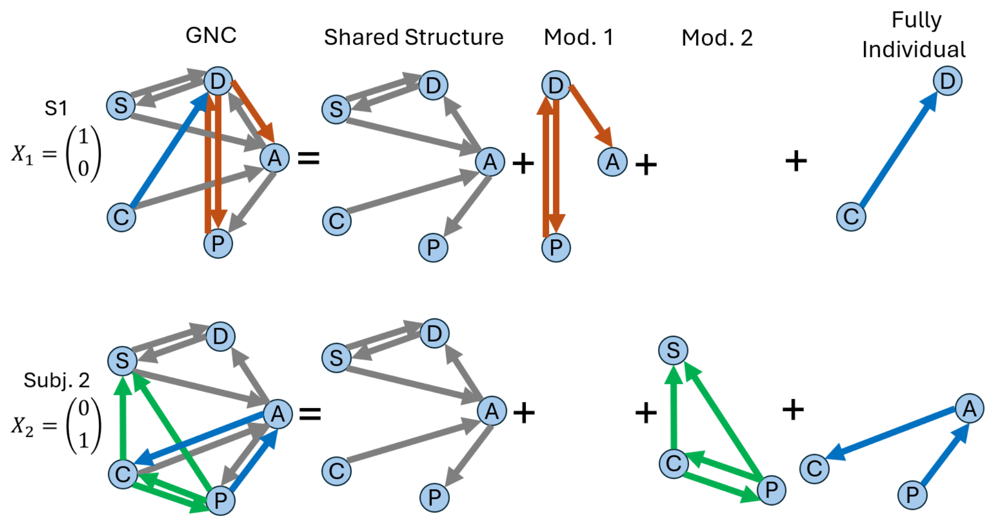


Figure 2: Illustration of the MOD-VAR framework with $p = 2$ binary moderators and fully individual terms $\{\Delta^{(Ik)}\}_{k\in[n]}$. This models scenarios where fully individual connections may be present across GCNs, in addition to shared and moderator GCN connections.

We illustrate the model introduced in (4) in Figure 2.

Our goal is to estimate the common component $\Phi^{(0)*}$, GCN moderator effects $\{\Phi^{(M\ell)*}\}_{\ell=1}^{p}$, and fully individual GCN connections $\{\boldsymbol{\Delta}^{(Ik)*}\}_{k=1}^{n}$.

3 Methodology

We introduce our method for simultaneously estimating the shared GCN connections across all subjects $\Phi^{(0)*}$, as well as the moderated effects $\{\Phi^{(M\ell)*}\}_{\ell=1}^{p}$ and fully individual connections $\{\boldsymbol{\Delta}^{(Ik)*}\}_{k=1}^{n}$. We achieve this via an ℓ_1-penalized time series estimation approach, which takes into account the estimation of all components. We first introduce the method when no individual remainders are present, *i.e.*, $\boldsymbol{\Delta}^{(I1)*} = \boldsymbol{\Delta}^{(I1)*} = \ldots = \boldsymbol{\Delta}^{(In)*} = \mathbf{0}$.

3.1 Mod-VAR With Only Moderator Components

Consider the case in which all inter-subject variability can be explained by our moderator variables, as described in (1). Then, our GCN follows the structure (4). Consider $\Phi =$

$[\Phi^{(0)}, \Phi^{(M1)}, \Phi^{(M2)}, \ldots, \Phi^{(Mp)}] \in \mathbb{R}^{d\times(d+dp)}$, and, for any vector of p moderators $\boldsymbol{x}$, we define $\Phi(\boldsymbol{x})$ as the $d \times d$ matrix given by

$$\Phi(\boldsymbol{x}) := \Phi^{(0)} + x_1\Phi^{(M1)} + x_2\Phi^{(M2)} + \ldots + x_p\Phi^{(Mp)}. \tag{5}$$

Our goal is to estimate $\hat{\Phi}$. Since this requires the estimation of a total of $q = (p+1)d^2$ parameters, the estimation can be challenging, specially in cases where the number of study subjects n or number of time points per subject $\{T_k\}_{k=1}^n$ are limited. In such scenarios, one may estimate Φ^* under the assumption of sparsity, *i.e.*, only a small number of parameters in Φ^* are non-zero, with the vast majority of the entries in Φ^* being zero. As seen in the literature, the estimation of such sparse VAR structures can be achieved through the use of ℓ_1-penalties [2; 7]. Here, we adapt such sparse-estimation techniques for detecting moderator effects in the VAR models of interest.

To derive a sparse estimation of Φ^*, we introduce the MOD-VAR optimization problem

$$\hat{\Phi} = \underset{[\Phi^{(0)},\Phi^{(M1)},\ldots,\Phi^{(Mp)}]}{\operatorname{argmin}} \frac{1}{2N}\sum_{k=1}^{n}\sum_{t=2}^{T_k}\|\boldsymbol{Y}_t^{(k)} - \Phi(\boldsymbol{X}_k)\boldsymbol{Y}_{t-1}^{(k)}\|_2^2 + \lambda_1\left\|\!\left|\Phi^{(0)}\right|\!\right\|_1 + \lambda_2\sum_{\ell=1}^{p}\left\|\!\left|\Phi^{(M\ell)}\right|\!\right\|_1. \tag{6}$$

where $N = T_1 + T_2 + \ldots + T_n - n$, and $\lambda_1, \lambda_2 > 0$ are tuning parameters. Observe that the optimization (6) aims to minimize the mean square forecasting error (MSFE) across all subjects. Furthermore, it incorporates a non-uniform penalty, that induces different levels of sparsity for the shared component $\Phi^{(0)}$ and the moderator effects $\{\Phi^{(M\ell)}\}_k$.

We can simplify the expression of (6) to a reduced matrix form. For this, consider $\mathbf{Z}^{(k)} = [\boldsymbol{Y}_T^{(k)}, \boldsymbol{Y}_{T-1}^{(k)}, \ldots, \boldsymbol{Y}_2^{(k)}]'$, $\mathbf{Y}^{(k)} = [\boldsymbol{Y}_{T-1}^{(k)}, \boldsymbol{Y}_{T-2}^{(k)}, \ldots, \boldsymbol{Y}_1^{(k)}]'$, and $\mathbf{E}^{(k)} = [\boldsymbol{\varepsilon}_T^{(k)}, \boldsymbol{\varepsilon}_{T-1}^{(k)} \ldots, \boldsymbol{\varepsilon}_2^{(k)}]'$. Here the matrices $\mathbf{Z}^{(k)}, \mathbf{Y}^{(k)}$ and $\mathbf{E}^{(k)}$ aggregate the contemporaneous, lagged and error terms for subject k, respectively. Furthermore, we introduce

$$\mathbf{W}^{(k)} := \left[\mathbf{Y}^{(k)}, X_{k1}\mathbf{Y}^{(k)}, \ldots, X_{kp}\mathbf{Y}^{(k)}\right] = \boldsymbol{X}_k' \otimes \mathbf{Y}^{(k)} \in \mathbb{R}^{(T_k-1)\times(pd+d)}.$$

One can show that $(\mathbf{Y}^{(k)})' \cdot \Phi^*(\boldsymbol{X}_k) = (\mathbf{W}^{(k)}) \cdot (\Phi^*)'$. We gather data from all subjects through the matrices $\mathbf{Z} := [\mathbf{Z}^{(1)'}, \mathbf{Z}^{(2)'}, \ldots, \mathbf{Z}^{(n)'}]'$, $\mathbf{W} := [\mathbf{W}^{(1)'}, \mathbf{W}^{(2)'}, \ldots, \mathbf{W}^{(n)'}]'$, and

$\mathbf{E} := [\mathbf{E}^{(1)'}, \mathbf{E}^{(2)'}, \dots, \mathbf{E}^{(n)'}]'$ We can summarize all relationships in (1) with the simplified matrix notation $\mathbf{Z} = \mathbf{W} \cdot (\Phi^*)' + \mathbf{E}$. The optimization problem (6) is equivalent to solving

$$\widehat{\Phi} := \underset{\Phi=[\Phi^{(0)},\Phi^{(M1)},\dots,\Phi^{(Mp)}]}{\operatorname{argmin}} \frac{1}{2N} \left|\left|\left| \mathbf{Z} - \mathbf{W}\Phi' \right|\right|\right|_F^2 + \lambda_1 \left|\left|\left| \Phi^{(0)} \right|\right|\right|_1 + \lambda_2 \left|\left|\left| [\Phi^{(M1)}, \dots, \Phi^{(Mp)}] \right|\right|\right|_1. \quad (7)$$

Thus, the estimation of $\hat{\Phi}$ can be reduced to a multiple regression optimization problem, with a non-uniform ℓ_1 penalty. We describe the optimization algorithm in Section 4.

3.2 Mod-VAR With Fully Individual Components

In many real data scenarios, moderator data may not be available, or the heterogeneity across subject may not be fully explained by the available moderator data. In such cases, we may aim to estimate additional fully individual connections in the GCN of each subject. We construct a set of artificial moderators to adapt optimization problem described in Section 3.1 for estimating common and fully individual components of the GCNs.

We consider the case in which we do not have moderator data available, and simply aim to estimate the shared and individual structure across multiple GCNs. This scenario is equivalent to the problem studied by the multi-VAR method [7]. For each subject k, we define the artificial moderator vector $\boldsymbol{X}_k^{(I)} = (0, \dots 0, , 1, 0, \dots, 0)'$, a column vector of size $n \times 1$, with zero in all entries except the k-th entry. Since $\boldsymbol{X}_k^{(I)}$ has n artificial moderators, the associated parameters of MOD-VAR model can be expressed in the form of a $d \times (nd+d)$ matrix $\Phi = [\Phi^{(0)}, \Phi^{(M1)}, \Phi^{(M2)}, \dots, \Phi^{(Mn)}]$. Observe that, since $\boldsymbol{X}_k^{(I)}$ is indicator of the k-th entry and (5), $\Phi(\boldsymbol{X}_k^{(I)}) = \Phi^{(0)} + \Phi^{(Mk)}$, for all $k \in [n]$. Since the moderator effect $\Phi^{(Mk)}$ appears only in the coefficients of subject k, $\Phi^{(Mk)}$ corresponds to a fully individual connections in the GCN for subject k. From this, we change our notation for the estimating parameter to $\Phi = [\Phi^{(0)}, \boldsymbol{\Delta}^{(I1)}, \boldsymbol{\Delta}^{(I2)}, \dots, \boldsymbol{\Delta}^{(In)}]$, where $\boldsymbol{\Delta}^{(Ik)}$ represents the fully individual GCN connections to subject k. In summary, the estimation of $\Phi = [\Phi^{(0)}, \boldsymbol{\Delta}^{(I1)}, \boldsymbol{\Delta}^{(I2)}, \dots, \boldsymbol{\Delta}^{(In)}]$

can be achieved by solving our proposed MOD-VAR optimization problem (7) with the artificial moderator design matrix $\mathbf{X}_{(I)} := [\boldsymbol{X}_1^{(I)}, \boldsymbol{X}_2^{(I)}, \ldots, \boldsymbol{X}_1^{(I)}]' = I_n \in \mathbb{R}^{n\times n}$.

We may also be interested in scenarios where the GCNs incorporates shared, moderator-based and fully individual connections across subjects, as described in Section 2.3. For this, consider, for n subjects, the $n\times p$ moderator data matrix $\mathbf{X}_{(M)} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n]' \in \mathbb{R}^{n\times p}$, and the artificial moderator matrix $\mathbf{X}_{(I)} \in \mathbb{R}^{n\times n}$. We construct the expanded matrix $\mathbf{X}_{(M,I)} := [\mathbf{X}_{(M)}, \mathbf{X}_{(I)}]$. By incorporating both data-driven and artificial moderators to our matrix $\mathbf{X}_{(M,I)}$, we can estimate all the parameters $\Phi^{(0)}$, $\{\Phi^{(M\ell)}\}_{\ell=1}^p$ and $\{\boldsymbol{\Delta}^{(Ik)}\}_{k=1}^n$ simultaneously by solving our proposed optimization problem (7).

4 Computational Algorithm and Estimation

We now discuss the procedure for optimizing (7), as well as for selecting the optimal tuning parameters λ_1 and λ_2 for fitting the MOD-VAR method.

4.1 Column-wise FISTA

The estimation of MOD-VAR requires solving the optimization problem (7). Since both terms $|||\mathbf{Z} - \mathbf{W}\Phi'|||_F^2/2N$ and $\mathcal{P}(\Phi) = \lambda_1 \left|\left|\left|\Phi^{(0)}\right|\right|\right|_1 + \lambda_2 \left|\left|\left|[\Phi^{(M1)}, \ldots, \Phi^{(Mp)}]\right|\right|\right|_1$ are decomposable by the rows of the matrix Φ, we can solve for Φ by solving for each row separately. Thus, for each $\ell \in [d]$ we aim to solve the optimization problem

$$\hat{\phi}_\ell = \underset{\phi\in\mathbb{R}^{d+dp}}{\operatorname{argmin}} \left\{ \frac{1}{2N} \|\mathbf{Z}_{\cdot\ell} - \mathbf{W}\phi\|_2^2 + \lambda_1 \|\phi_{[1:d]}\|_1 + \lambda_2 \|\phi_{[(d+1):(d+dp)]}\|_1 \right\}. \tag{8}$$

Once we obtain $\hat{\phi}_1, \hat{\phi}_2, \ldots, \hat{\phi}_d \in \mathbb{R}^d$, we construct $\hat{\Phi} = [\hat{\phi}_1, \hat{\phi}_2, \ldots, \hat{\phi}_d]' \in \mathbb{R}^{d\times(d+dp)}$. Then, the estimated common portion of the GCN is defined as $\hat{\Phi}^{(0)} = \hat{\Phi}_{\cdot,[1:d]}$, and, for any $k \in [p]$ the estimated effect of the k-th moderator on the GCN defined as $\hat{\Phi}^{(Mk)} =$

$\hat{\Phi}_{\cdot,[(d+d(k-1)):(d+dk)]}$. A summary of this procedure is provided in Algorithm 1. As discussed in Section 3.2, fully individual components can be incorporated into the model via artificial moderator variables. Therefore, the procedure described in Algorithm 1 can be applied for estimating shared, moderated and fully individual connections across GCNs, by simple modifications of the moderator design matrix $\mathbf{X}$.

Algorithm 1: MOD-VAR.

Input: Time series $\{\boldsymbol{Y}_t^{(k)} : k \in [n],\, t \in [T_k]\} \subset \mathbb{R}^d$, moderators $\mathbf{X} \in \mathbb{R}^{n\times p}$, tuning parameters $\lambda_1, \lambda_2 > 0$.

for $k \in [n]$ **do**

  Define $\mathbf{Z}^{(k)} = [\boldsymbol{Y}_2^{(k)}, \boldsymbol{Y}_3^{(k)}, \ldots, \boldsymbol{Y}_{T_k}^{(k)}]' \in \mathbb{R}^{(T_k-1)\times d}$.

  Define $\mathbf{Y}^{(k)} = [\boldsymbol{Y}_1^{(k)}, \boldsymbol{Y}_2^{(k)}, \ldots, \boldsymbol{Y}_{T_k-1}^{(k)}]' \in \mathbb{R}^{(T_k-1)\times d}$.

  Set $\boldsymbol{X}_k^* := (1, \mathbf{X}_{k\cdot}')' \in \mathbb{R}^{p+1}$.

  Define $\mathbf{W}^{(k)} = \boldsymbol{X}_k^* \otimes \mathbf{Y}^{(k)} = [\mathbf{Y}^{(k)}, X_{k1}\mathbf{Y}^{(k)}, \ldots, X_{kp}\mathbf{Y}^{(k)}] \in \mathbb{R}^{(T_k-1)\times(d+dp)}$.

end

Define $\mathbf{Z} = [\mathbf{Z}^{(1)\prime}, \mathbf{Z}^{(2)\prime}, \ldots, \mathbf{Z}^{(n)\prime}]' \in \mathbb{R}^{N\times d}$.

Define $\mathbf{W} = [\mathbf{W}^{(1)\prime}, \mathbf{W}^{(2)\prime}, \ldots, \mathbf{W}^{(n)\prime}]' \in \mathbb{R}^{N\times(d+dp)}$.

for $\ell \in [d]$ **do**

  Solve $\hat{\boldsymbol{\phi}}_\ell = \operatorname{argmin}_{\boldsymbol{\phi}\in\mathbb{R}^{d(p+1)}} \|\mathbf{Z}_{\cdot\ell} - \mathbf{W}\boldsymbol{\phi}\|_2^2/2N + \lambda_1\|\boldsymbol{\phi}_{1:d}\|_1 + \lambda_2\|\boldsymbol{\phi}_{(d+1):(d+pd)}\|_1$.

end

Define $\hat{\Phi} := \hat{\Phi}(\lambda_1,\lambda_2) = [\hat{\boldsymbol{\phi}}_1, \hat{\boldsymbol{\phi}}_2, \ldots, \hat{\boldsymbol{\phi}}_d]' \in \mathbb{R}^{d\times(d+pd)}$.

Set $\hat{\Phi}^{(0)} := \hat{\Phi}^{(0)}(\lambda_1,\lambda_2) = \hat{\Phi}_{\cdot,[1:d]}$.

For $s \in [p]$, define $\hat{\Phi}^{(Ms)} := \hat{\Phi}^{(Ms)}(\lambda_1,\lambda_2) = \hat{\Phi}_{\cdot,[(d+d(s-1)):(d+ds)]}$.

Output: Common effects $\hat{\Phi}^{(0)}$, and moderated effects $\{\hat{\Phi}^{(Ms)}\}_{s=1}^p$

In order to solve the optimization problem (8) for each $\ell \in [d]$, we apply the fast iterative shrinkage-threshold algorithm (FISTA) algorithm [4]. For each $\ell \in [d]$, we define $\mathcal{L}^{(\ell)}(\phi) = -\boldsymbol{\phi}'\mathbf{W}'\mathbf{W}\boldsymbol{\phi}/2N - \mathbf{Z}_{\cdot\ell}'\mathbf{W}\boldsymbol{\phi}/N$, and the penalty term $\mathcal{P}(\boldsymbol{\phi}) = \lambda_1\|\boldsymbol{\phi}_{1:d}\|_1 + \lambda_2\|\boldsymbol{\phi}_{(d+1):(d+pd)}\|_1$. It can be shown that optimizing (8) is equivalent to solving the optimization problem $\operatorname{argmin}_{\boldsymbol{\phi}} \mathcal{L}^{(\ell)}(\boldsymbol{\phi}) + \mathcal{P}(\boldsymbol{\phi})$. Furthermore, FISTA requires calculating a Lipschitz constant $L > 0$ for the gradient $\nabla\mathcal{L}^{(\ell)}(\boldsymbol{\phi}) = (\mathbf{W}'\mathbf{W}/N)\phi - \mathbf{W}'\mathbf{Z}_{\cdot\ell}/N$. The inputs of our implementation of the FISTA algorithm applied to each $\ell \in [d]$ are derived from the matrices $\mathbf{M} := \mathbf{W}'\mathbf{W}/N$, $\boldsymbol{b}_\ell := \mathbf{W}'\mathbf{Z}_{\cdot\ell}$, the tuning parameters $\lambda_1, \lambda_2 > 0$, the Lipschitz constant $L := \|\!|\mathbf{W}'\mathbf{W}|\!\|_2/2N$, and a maximum iteration number I, and a tolerance level $\varepsilon > 0$. In addition, we may improve efficiency of our estimation by utilizing a warm start $\boldsymbol{x}_0 \in \mathbb{R}^{d+dp}$.

Our FISTA implementation consists of the following steps. We start by setting $s_1 = 1$, and initializing $\boldsymbol{x}_0$ if a warm start is not provided. Then, we set $\boldsymbol{y}_1 = \boldsymbol{x}_0$, and define the vector of penalization weights as $\boldsymbol{w} = (\lambda_1 \mathbb{1}'_d, \lambda_2 \mathbb{1}'_{dp})' \in \mathbb{R}^{d+dp}$. Then, for any $i = 1, 2, \ldots, I$, the updates for $\boldsymbol{x}_i$, s_i and $\boldsymbol{y}_{i+1}$ are given by the equations

$$\begin{aligned}
\boldsymbol{x}_i &:= ST_{\boldsymbol{w}/L}(\boldsymbol{y}_i - \nabla \mathcal{L}^{(\ell)}(\boldsymbol{y}_i)/L), \\
s_{i+1} &:= (1 + \sqrt{1 + 4s_i^2})/2, \\
\boldsymbol{y}_{i+1} &:= \boldsymbol{x}_i + ((s_i + 1)/s_{i+1})(\boldsymbol{x}_i - \boldsymbol{x}_{i-1}),
\end{aligned}$$

where, for any $\boldsymbol{w}, \boldsymbol{a} \in \mathbb{R}^m$, we define $ST_{\boldsymbol{w}}(\boldsymbol{a}) \in \mathbb{R}^m$ as the weighted entrywise soft thresholding function given by the entrywise value $[ST_{\boldsymbol{w}}(\boldsymbol{a})]_i = sign(a_i) \cdot \max\{0, |a_i| - |w_i|\}$. The full procedure is summarized in Algorithm 2.

Algorithm 2: FISTA for ℓ-step of MOD-VAR estimation.

Input: $\mathbf{M} := \mathbf{W}'\mathbf{W}/N \in \mathbb{R}^{(d+dp)\times(d+dp)}$, $\boldsymbol{b}_\ell := \mathbf{W}'\mathbf{Z}_{\cdot\ell}/N \in \mathbb{R}^{d+dp}$, hyperparameters $\lambda_1, \lambda_2 > 0$, Lipschitz constant $L := |||\mathbf{M}|||_2$, maximum iterations $I > 0$, tolerance $\varepsilon > 0$, warm start $\boldsymbol{x}_0 \in \mathbb{R}^{d+dp}$.

Initialize $s_1 = 1$, $\boldsymbol{y}_1 = \boldsymbol{x}_0$, and $\boldsymbol{x}_1 = ST_{\boldsymbol{w}/L}(\boldsymbol{y}_1 - \nabla \mathcal{L}^{(\ell)}(\boldsymbol{y}_1)/L)$.

Set $\boldsymbol{w} = (\lambda_1 \mathbb{1}'_d, \lambda_1 \mathbb{1}'_{dp})' \in \mathbb{R}^{d+dp}$ and $i = 1$.

while $i \leq I$ *and* $\|\boldsymbol{x}_i - \boldsymbol{x}_{i-1}\|_2 > \varepsilon$ **do**

- Calculate $\nabla \mathcal{L}^{(\ell)}(\boldsymbol{y}_i) = \mathbf{M}\boldsymbol{y}_i - \boldsymbol{b}_\ell$.
- Set $\boldsymbol{x}_i := ST_{\boldsymbol{w}/L}(\boldsymbol{y}_i - \nabla \mathcal{L}^{(\ell)}(\boldsymbol{y}_i)/L)$.
- Set $s_{i+1} = (1 + \sqrt{1 + 4s_i^2})/2$.
- Set $\boldsymbol{y}_{i+1} = \boldsymbol{x}_i + ((s_i + 1)/s_{i+1})(\boldsymbol{x}_i - \boldsymbol{x}_{i-1})$.
- Set $i \leftarrow i + 1$.

end

Set $\hat{\boldsymbol{\phi}}_\ell := \hat{\boldsymbol{\phi}}_\ell(\lambda_1, \lambda_2) = \boldsymbol{x}_{i-1}$.

4.2 Tuning Parameter Selection

To select the optimal MOD-VAR fit, we must select the optimal $\lambda_1, \lambda_2 > 0$ over a range of potential values. We describe three strategies for the selection of $\lambda_1, \lambda_2 > 0$: by-subject cross validation, rolling window and cross validation, and an adaptive version of our proposed moderator vector auto-regressive model (adaMOD-VAR).

4.2.1 By Subject Cross-Validation

We introduce the by-subject cross-validation (BSCV) procedure for fitting our proposed MOD-VAR method. Our J-fold BSCV procedure is as follows. First, we divide our set of subjects $[n]$ into J approximately equally sized partitions $\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_J$. Then, for each $j \in [J]$, and choice of tuning parameters $\lambda_1 > 0$, $\lambda_2 > 0$, we fit $\hat{\Phi}_{(BSCV,j)}(\lambda_1, \lambda_2)$ as the solution to (6) based exclusively on the data of the subjects $[n] \setminus \mathcal{C}_j$. This provides us with the common GCN structure $\hat{\Phi}^{(0)}_{(BSCV,j)}(\lambda_1, \lambda_2) \in \mathbb{R}^{d\times d}$ and for each $s \in [p]$, the effect of the s-moderator onto the GCN $\{\hat{\Phi}^{(M\ell)}_{(BSCV,j)}(\lambda_1, \lambda_2)\}_{\ell=1}^{p} \subset \mathbb{R}^{d\times d}$. Then, for each subject $k \in \mathcal{C}_j$, we define the cross-validation predicted GCN based on the moderator vector $\boldsymbol{X}_k$,

$$\hat{\Phi}^{(\lambda_1,\lambda_2)}_{(BSCV,j)}(\boldsymbol{X}_k) = \hat{\Phi}^{(0)}_{(BSCV,j)}(\lambda_1, \lambda_2) + \sum_{s=1}^{p} X_{ks}\hat{\Phi}^{(Ms)}_{(BSCV,j)}(\lambda_1, \lambda_2) \in \mathbb{R}^{d\times d}.$$

Here, $\hat{\Phi}^{(\lambda_1,\lambda_2)}_{(BSCV,j)}(\boldsymbol{X}_k)$ provides an estimation of the GCN corresponding to subject $k \in \mathcal{C}_j$, with no use of the data of subject k, instead relying only on the for the subject set $[n] \setminus \mathcal{C}_j$. Finally, we define the by-subject cross-validation mean-squared forecasting error as

$$\begin{aligned} BSCV.MSFE(\lambda_1, \lambda_2) &= \frac{1}{J}\sum_{j\in[J]}\sum_{k\in\mathcal{C}_j}\frac{1}{T_k - 1}\left\|\mathbf{Z}^{(k)} - \mathbf{Y}^{(k)}\hat{\Phi}^{(\lambda_1,\lambda_2)}_{(BSCV,j)}(\boldsymbol{X}_k)\right\|_F^2. \\ &= \frac{1}{J}\sum_{j\in[J]}\sum_{k\in\mathcal{C}_j}\frac{1}{T_k - 1}\sum_{t=2}^{T_k}\|\boldsymbol{Y}^{(k)}_t - \hat{\Phi}^{(\lambda_1,\lambda_2)}_{(BSCV,j)}(\boldsymbol{X}_k)\boldsymbol{Y}^{(k)}_{t-1}\|_2^2 \end{aligned}$$

We select the pair $\lambda_1, \lambda_2 > 0$ that minimize the $BSCV.MSFE(\lambda_1, \lambda_2)$.

This approach enjoys major computational advantages compared to other procedures like rolling-window cross-validation, since it only requires the fit of J runs of our MOD-VAR fitting procedure. We emphasize that by-subject cross validation can be used exclusively when no fully individual connections are expected across models.

4.2.2 Rolling Window Cross-Validation

Rolling window cross-validation is an alternative procedure for the selection of an optimal MOD-VAR model, based on the method proposed in [22]. First, we select $T = \min_{k\in[n]}\{T_k - 1\}$, and we set $J = \lfloor T/3 \rfloor$. Then, for any $j \in [J]$, we select the observations $\cup_{k\in[n]}\{Y_t^{(k)} : t \in [T_k - j - 2]\}$ as the training set. With this, we obtain the by-subject GCN estimators $\hat{\Phi}^{(k;\lambda_1,\lambda_2)}_{(RWCV,j)} \in \mathbb{R}^{d\times d}$ for each potential choice of $\lambda_1, \lambda_2 > 0$. Finally, we measure the rolling window cross validation mean squared forecasting error, given by,

$$RWCV.MSFE(\lambda_1, \lambda_2) = \frac{1}{J}\sum_{j\in[J]}\frac{1}{n}\sum_{k\in[n]}\|\mathbf{Y}^{(k)}_{T_k-j-1} - \hat{\Phi}^{(k;\lambda_1,\lambda_2)}_{(RWCV,j)} \cdot \mathbf{Y}^{(k)}_{T_k-j}\|_2^2.$$

We select the values $\lambda_1^*, \lambda_2^* > 0$ with the lowest RWCV.MSFE value.

4.2.3 Adaptative MOD-VAR

To improve upon the performance of our original estimator, we propose a refinement of our MOD-VAR estimator through an adaptive approach [36]. For this, we consider $\hat{\Phi} = [\hat{\Phi}^{(0)}, \hat{\Phi}^{(M1)}, \ldots, \hat{\Phi}^{(Mp)}, \hat{\mathbf{\Delta}}^{(I1)}, \ldots, \hat{\mathbf{\Delta}}^{(In)}] \in \mathbb{R}^{d\times(d+dp+dn)}$ be an initial estimator of our proposed MOD-VAR. Then, given a constant $\gamma > 0$ we define the adaptive weight for the shared and moderated coefficients $[\hat{\Phi}^{(0)}, \hat{\Phi}^{(M1)}, \ldots, \hat{\Phi}^{(Mp)}]$ as,

$$\omega_{ada}(\hat{\Phi})_{ij} = |\hat{\Phi}_{ij}|^{-\gamma}, \qquad \forall\, 1 \le i \le d,\, 1 \le j \le p+1.$$

For the fully individual connectivity portion $[\hat{\mathbf{\Delta}}^{(I1)}, \ldots, \hat{\mathbf{\Delta}}^{(In)}]$, we select our adaptive penalty similar to the proposal of Fisher et al. [8]. First, based on $\hat{\Phi}$, we calculate the by-subject GCNs, $\hat{\Phi}^{(1)}, \hat{\Phi}^{(2)}, \ldots, \hat{\Phi}^{(n)} \in \mathbb{R}^{d\times d}$. Then, we calculate the entrywise median

GCN $\tilde{\Phi} \in \mathbb{R}^{d\times d}$, where $\tilde{\Phi}_{ij}$ is the median of $\hat{\Phi}_{ij}^{(1)}, \hat{\Phi}_{ij}^{(2)}, \ldots, \hat{\Phi}_{ij}^{(n)}$. Finally, we define the penalty adaptive weight for $\mathbf{\Delta}_{ij}^{(Ik)}$ as,

$$\omega_{ada}(\hat{\Phi})_{i,d(p+k)+j} = |\Delta_{ij}^{(Ik)} - \tilde{\Phi}_{ij}|^{-\gamma}, \qquad \forall\, 1 \le i,j \le d,\, 1 \le k \le n.$$

From this, we define the adaptively-weighted ℓ_1 penalty for Φ as,

$$P_{ada}(\Phi) = \sum_{\ell=1}^{p} \sum_{i,j=1}^{d} \omega_{ada}(\hat{\Phi})_{ij} \cdot |\Phi_{ij}|. \tag{9}$$

Then, our adaptive MOD-VAR estimator is given as a solution to the optimization problem

$$\widehat{\Phi}_{ada}(\lambda) = \operatorname*{argmin}_{\Phi\in\mathbb{R}^{d\times(d+dp))}} \frac{1}{2N} |||\mathbf{Z} - \mathbf{W}\Phi|||_F^2 + \lambda P_{ada}(\Phi), \qquad \textit{for } \lambda > 0.$$

As observed in our numerical simulations, the adaptive procedure has significant improvements in terms of both forecasting, consistency, and edge-recovery performance.

5 Theoretical Properties

We explore the theoretical properties of our proposed MOD-VAR, and provide high-dimensional consistency guarantees of estimation. In particular, we are interested in the estimation error $\left|\left|\left|\hat{\Phi} - \Phi^*\right|\right|\right|_F$ and ℓ_1-error $\|\hat{\Phi} - \Phi^*\|_1$. To explore this, consider the MOD-VAR matrix of true parameters Φ^* is s-sparse: $\|\Phi^*\|_0 \le s$. To simplify our exposition, we assume that all time series have the same number of time observations, *i.e.*, $T := T_1 = T_2 = \ldots = T_n$. Then, we denote N the total number of observations available as $N = n(T-1)$.

To establish our guarantees, consider the following set of assumptions.

Assumption 1 (Bounded moderators). *Assume the moderator vectors $\boldsymbol{X}, \boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ are independent, identically distributed and entrywise bounded random vectors i. e., $|X_i| \le 1$, with mean zero, and covariance matrix $\Sigma_X = \mathbb{E}[\boldsymbol{X}\boldsymbol{X}']$. Furthermore, assume that for some constant $0 < \eta < 1$, we have that $\sup_{\|x\|_\infty \le 1} |||\Phi^*(\boldsymbol{x})|||_2 < \eta < 1$*

Assumption 2 (Curvature condition)**.** *Let* $\Lambda_Y := \inf_{\|\boldsymbol{x}\|_\infty \le 1} \Lambda_{\min}(Cov(Y_t^k|X_k))$*, and* $\alpha = \Lambda_Y \cdot \Lambda_{\min}(\Sigma_X)/2$*. There exists a universal constant* $C > 0$ *such that, for some* $t > 0$*,*

$$\alpha > Cs^2 \left(\sqrt{\frac{\log(d^2 p) + t}{n}} + \frac{\log(2n)(\log(pd^2) + t)}{n} \right).$$

Our Assumption 1 requires that the moderator variables satisfy the entrywise bound $|X_i| \le 1$. This is not restrictive, since in applications, covariates often correspond to categorical or bounded score variables. For example, in the analysis of behavioral data from longitudinal studies, moderators of interest are often variables such as socioeconomic status, educational status, and age, which can be coded and normalized as discrete, centered, and bounded variables. The condition $\sup_{\|x\|_\infty \le 1} |||\Phi^*(\boldsymbol{x})|||_2 < \eta < 1$ serves to ensure that all the sampled time series are stable [2].

The convergence of our MOD-VAR estimator relies on ensuring sufficient curvature of the matrix $\mathbf{W}'\mathbf{W}/N \in \mathbb{R}^{(d+dp)\times(d+dp)}$. Since the design matrix $\mathbf{W}$ is composed of the blocks $\boldsymbol{X}_k^* \otimes \mathbf{Y}^{(k)}$, where $\boldsymbol{X}_k^* = (1, \boldsymbol{X}_k')' \in \mathbb{R}^{p+1}$, the minimum eigenvalue of $\mathbb{E}[\mathbf{W}'\mathbf{W}/N]$ can be controlled by imposing minimum eigenvalue conditions of Σ_X and the conditional covariances of $Cov(\boldsymbol{Y}_t^{(k)}|\boldsymbol{X}_k)$. Similar minimum eigenvalue controls are commonly used in the literature of high-dimensional regression literature [19; 2; 30].

Under Assumptions 1 and 2, we derive high-dimensional guarantees of convergence.

Theorem 1. *Consider multiple subject data* $\{(\mathbf{Y}_k, \boldsymbol{X}_k)\}_{k=1}^n$ *that follows the MOD-VAR model* (4)*. Furthermore, suppose Assumption 1 holds, Assumption 2 is satisfied for a given* $t > 0$*, and* $n > 16\log(2n)\log(pd^2)$*. Let* $C_M := \Lambda_{\max}(\Sigma_\varepsilon)(2 + 4\eta/(1-\eta)^2)$*. Then, assuming that* $\lambda := \lambda_1 = \lambda_2$ *satisfies the inequality* $\lambda > 16\pi C_M \sqrt{\log(pd^2)/N}$*, we have that, with* *probability at least* $1 - 3e^{-t} - (6/(pd^2))^2$,

$$\|\hat{\Phi} - \Phi^*\|_2 \leq \frac{12\sqrt{s}\pi C_M}{\alpha}\sqrt{\frac{\log(pd^2)}{N}}; \tag{10}$$

$$\|\hat{\Phi} - \Phi^*\|_1 \leq \frac{72 s\pi C_M}{\alpha}\sqrt{\frac{\log(pd^2)}{N}}. \tag{11}$$

Our Theorem 1 provides guarantees on the convergence of our proposed MOD-VAR, even in high dimensional scenarios. First, notice that our bound on $\|\hat{\Phi} - \Phi^*\|_2$ can be translated into a bound on the individual GCN estimation,

$$\begin{aligned}\max_{k=1,\ldots,n} \|\hat{\Phi}(\boldsymbol{X}_k) - \Phi^*(\boldsymbol{X}_k)\|_1 &\leq \|\hat{\Phi}^{(0)} - \hat{\Phi}^{(0)*}\|_1 + \max_{k=1,\ldots,n}\sum_{i=1}^{p}|X_{ki}|\|\hat{\Phi}^{(Mi)} - \hat{\Phi}^{(Mi)*}\|_1 \\ &\leq \|\hat{\Phi} - \Phi^*\|_1\end{aligned}$$

As a consequence of this, as long as $\log(pd^2) = O(N)$, we may ensure the ℓ_1-norm consistency of our proposed MOD-VAR estimator. The rate $O(\sqrt{(\log(p) + 2\log(d))/N})$ described in (11) depends on the total sample size $N = \sum_{i=1}^{n}(T_i - 1)$. As long as $\log(p)$ is moderate (for example, if $p = O(d)$), the rate obtained in Theorem 1 represents a significant improvement to the rate of $O(\sqrt{2\log(d)/T})$ attained by the individual VAR estimation without moderators. This demonstrates the significant improvement of performing joint estimation when moderator effects are present in our time series model. Furthermore, notice that since the convergence depends on N and not on T, consistent estimation is possible, even in cases where the number of available observations per subject T is limited, as long as n the number of subjects n is large. This is possible thanks to the knowledge transfer across GCN's facilitated by the moderator data. The main requirement on the number of subjects n is $\log(pd^2) = o(n/\log(n))$, found in Assumption 2. This ensures that there is sufficient heterogeneity in the moderator data, and consistent estimation of Σ_X, which in turn facilitates the estimation of the GCN moderator effects.

6 Numerical Simulations

To assess the performance of our proposed MOD-VAR, and compare it to methods in the literature, we perform a set of numerical simulation experiments.

6.1 Data Generating Method

To generate our data, we select a dimension $d \in \{10, 20\}$, number of available moderator variables $p \in \{2, 5\}$, number of available time series $n \in \{15, 20, 25\}$, and number of available observations per time series $T \in \{30, 60, 90\}$.

To generate our data, we generate a matrix of moderator variables $\mathbf{X} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots \boldsymbol{X}_n] \in \mathbb{R}^{n \times p}$, as well as matrices of shared GCN connections $\Phi^{(0)*} \in \mathbb{R}^{d \times d}$, the GCN moderator effects $\{\Phi^{(Ms)*}\}_{s=1}^{p} \subset \mathbb{R}^{d \times d}$, and the matrices of fully individual connections $\{\Delta^{(Ik)*}\}_{k=1}^{n} \subset \mathbb{R}^{d \times d}$. Once these matrices are generated, for each $k \in [n]$, we define the GCN for the k-th time series based on the moderators $\boldsymbol{X}_k = (X_{k1}, \ldots X_{kp})'$ as,

$$\Phi^{(k)*} = \Phi^{(0)*} + \sum_{s=1}^{p} X_{ks} \Phi^{(Ms)*} + \Delta^{(Ik)*} \in \mathbb{R}^{d \times d}.$$

Then, for each $k \in [n]$, we generate the time series data $\{\boldsymbol{Y}_t^{(k)}\}_{t=1}^{T+400}$, by first setting $\boldsymbol{Y}_0^{(k)} = \mathbf{0}$, and $\boldsymbol{Y}_t^{(k)} = \Phi^{(k)*} \boldsymbol{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t$, where $\{\boldsymbol{\varepsilon}_t\}_{t=1}^{T+403} \sim N_d(\mathbf{0}, (0.1) \cdot \mathrm{I}_d)$. Then, we discard the first 400 time points of each time series. The additional 3 time points will be used to evaluate out-of-sample forecasting performance.

The simulation of our data requires a procedure for generating the matrix of moderator variables $\mathbf{X} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots \boldsymbol{X}_n]' \in \mathbb{R}^{n \times p}$, and the GNC matrices $\Phi^{(0)*}$, $\{\Phi^{(Ms)*}\}_{s=1}^{p}$ and $\{\Delta^{(Ik)}\}_{k=1}^{n}$. To generate $\mathbf{X} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots \boldsymbol{X}_n]' \in \mathbb{R}^{n \times p}$. we set $\boldsymbol{X}_k = (X_{k1}, \ldots, X_{kp}) \overset{i.i.d.}{\sim} (0.25) \cdot \delta_0 + (0.75) \cdot Unif([-1, -0.5] \cup [0.5, 1])$. To generate the GNC matrices, we introduce $q_0, q_m, q_\Delta \in [0, 1]$ the probabilities of shared, moderated and fully individual connection probabilities. With these parameters in place, we generate our data as follows. First, we generate the shared network $\Phi^{(0)*} \in \mathbb{R}^{d\times d}$, moderated networks $\{\Phi^{(Ms)*}\}_{s=1}^{p} \subset \mathbb{R}^{d\times d}$, and fully individual network connections $\{\Delta^{(Ik)*}\}_{k=1}^{n} \subset \mathbb{R}^{d\times d}$ as follows:

$$\{\Phi_{ij}^{(0)*}\}_{i,j=1}^{d} \overset{i.i.d.}{\sim} (1-q_0)\delta_0 + q_0 Unif([-0.9,-0.5] \cup [0.5, 0.9]);$$

$$\{\Phi_{ij}^{(Ms)*}\}_{i,j=1}^{d} \overset{i.i.d.}{\sim} (1-q_m)\delta_0 + q_m Unif([-0.9,-0.5] \cup [0.5, 0.9]);$$

$$\{\Delta_{ij}^{(Ik)*}\}_{i,j=1}^{d} \overset{i.i.d.}{\sim} (1-q_\Delta)\delta_0 + q_\Delta Unif([-0.9,-0.5] \cup [0.5, 0.9]).$$

Thus, the parameters q_0, q_m and q_Δ control the level of sparsity of the matrices $\Phi^{(0)*}$, $\{\Phi^{(Ms)*}\}_{s=1}^{p}$ and $\{\Delta^{(Ik)*}\}_{k=1}^{n}$. In particular, notice that if $q_m = 0$, it follows that $\Phi^{(Ms)*} = 0$ for all $s \in [p]$. Thus, $q_m = 0$ corresponds to a data generating scenario in which the moderator variables do not have any effect on the connectivity pattern across the GCNs. Furthermore, $q_\Delta = 0$ implies $\Delta^{(Ik)} = \mathbf{0}$ for all $k \in [n]$, which corresponds to an absence of fully individual structure in our time series. In our simulations, we explore different choices of q_0, q_m and q_Δ to explore the impact that moderator effects, fully individual effects and the absence of each have on the GNC estimation accuracy.

6.2 Methods and Performance Metrics

We compare our proposed MOD-VAR to methods in the literature of multiple time series modeling. First, we consider the vector auto-regressive (VAR) model [2] applied separately to each time series. This method is implemented via the `bigVAR` R package [21]. We also compare to the original and adaptive multi-VAR methods [7], implemented via the `multivar` package. Finally, we compare our MOD-VAR method, fitted with rolling-window cross-validation, BIC and adaptive procedures.

To evaluate the performance, we consider the mean ℓ_1 and Frobenius GNC estimation errors, and root mean squared forecasting errors, given by

$$E_{\ell_1}(\widehat{\Phi}, \Phi^*) = \frac{1}{n}\sum_{k=1}^{n} \|\widehat{\Phi}^{(k)} - \Phi^{(k)*}\|_1,$$

$$E_F(\widehat{\Phi}, \Phi^*) = \frac{1}{n}\sum_{k=1}^{n} \left\|\widehat{\Phi}^{(k)} - \Phi^{(k)*}\right\|_F,$$

$$RMSFE(\widehat{\Phi}, \Phi^*) = \left(\frac{1}{n}\sum_{k=1}^{n} \frac{1}{3}\sum_{h=1}^{3} \|\mathbf{Y}^{(k)}_{T+h-1} - \widehat{\Phi}^{(k)}\mathbf{Y}^{(k)}_{T+h}\|_2^2\right)^{1/2},$$

respectively. In our supplementary materials, we incorporate results on mean true positive rate of connection, mean false positive rate, and computational time.

6.3 Results: Only Moderator-Driven Heterogeneity

First, we explore the scenario in which all the heterogeneity across the time series models can be accounted by the p moderator variables. This corresponds to the case $q_\Delta = 0$, *i.e.,* $\Delta^{(Ik)} = 0$ for all $k \in [n]$. For this case, we explore the cases $(q_0, q_m) \in \{(0.075, 0.025), (0.025, 0.075)\}$. We provide visualizations for the E_{ℓ_1}, E_F and $MSFE$ for $d = 10$ in this section. We provide visualizations for the TPR, FPR in our Supplementary Materials.

As we observe in Figure 3, both the MOD-VAR and adaptive MOD-VAR attain the best performance in terms of ℓ_1-norm error. Notice that, depending on the degree of commonality among the GCNs, the second best estimation may be attained by the individual VAR estimation or the Multi-VAR method. In the scenario with the least cross-subject heterogeneity in which 75% of the GCN is shared, and $p = 2$, we have that the Multi-VAR method is the next best performing method. On the other hand, in scenarios with high heterogeneity where only 25% of entries are shared and $p = 5$, we have that the VAR method is the second best performing method. Our proposed MOD-VAR attains a significantly better performance, since it can model joint structure, and uses moderator data to drastically reduce the parameter complexity of estimating the heterogeneous structure.

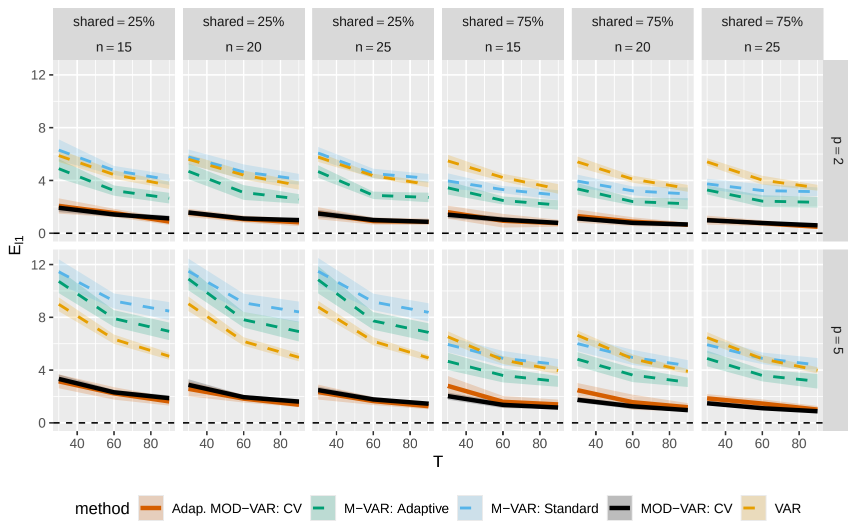


Figure 3: Visualization of the mean ℓ_1 parameter estimation error for our proposed MOD-VAR, as well as methods in the literature. for data where the heterogeneity across subjects is fully explained by moderator variables. Our proposed MOD-VAR and adaptive MOD-VAR attain the best performance, for all choices of number of subjects n, moderators p and percentage of shared structure in the GCN.

6.4 Results: Moderator and Fully Individual Heterogeneity

To further explore the performance of our estimation procedure, accounting for the potential presence of both moderator effects, as well as fully individual connections in the GNC of some individuals, we perform an additional simulation scenario. For this, we set $(q_0, q_m) \in \{(0.075, 0.025), (0.025, 0.075)\}$ and $q_\Delta = 0.03$. We provide visualizations for the E_{ℓ_1}, E_F and $MSFE$ for $d = 10$ in this section. We provide visualizations for the TPR, FPR in our Supplementary Materials. add interpretation of the results.

7 Analysis of fMRI Time-Series Data

We analyze a multi-subject time series data of resting state fMRI reported in Hwang et al. [12]. Critically, this dataset contains intensive measurements of 9 resting state fMRI measurements for $n = 26$ subjects over $T = 150$ observations per subject. The dataset consists of measurements of brain activity for regions composing the Frontoparietal Net-

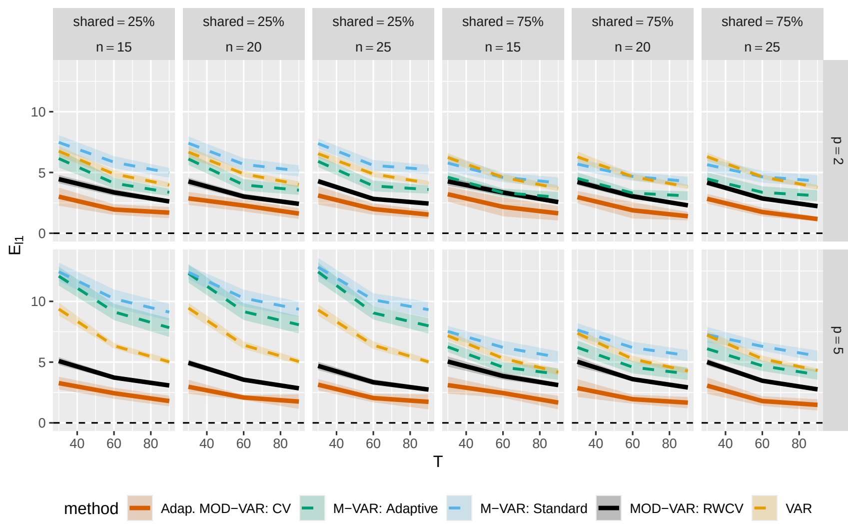


Figure 4: Visualization of the mean ℓ_1 parameter estimation error for our proposed MOD-VAR, as well as methods in the literature the subject heterogeneity is a combination of moderator effects and fully individual connections. Our proposed adaptive MOD-VAR attain the best performance, for all choices of number of subjects n, moderators p and percentage of shared structure in the GCN.

work (dorsolateral prefrontal cortex, posterior parietal cortex, dorsal anterior cingulate cortex/pre-supplementary motor area), default mode network (medial prefrontal cortex, posterior cingulate cortex/precuneus, inferior parietal lobule, medial temporal lobe), and salience network (anterior insula, anterior cingulate cortex). In addition, this dataset contains 4 time-invariant moderators, including participant age, biological sex, locus coeruleus volume (LC volume), and locus coeruleus contrast-to-noise ratio. It has been proposed and demonstrated that both age [1; 10] and locus coeruleus structure [34; 27] are associated with activity within these different functional brain networks.

We compare the out of sample forecasting accuracy between VAR, Multi-VAR, and MOD-VAR models on this fMRI dataset through the following experiment. For each method, we train each model on the data corresponding to the first T_{train} observations per subject, where $10 \leq T_{train} \leq 100$, resulting in the estimated by-subject auto-regressive

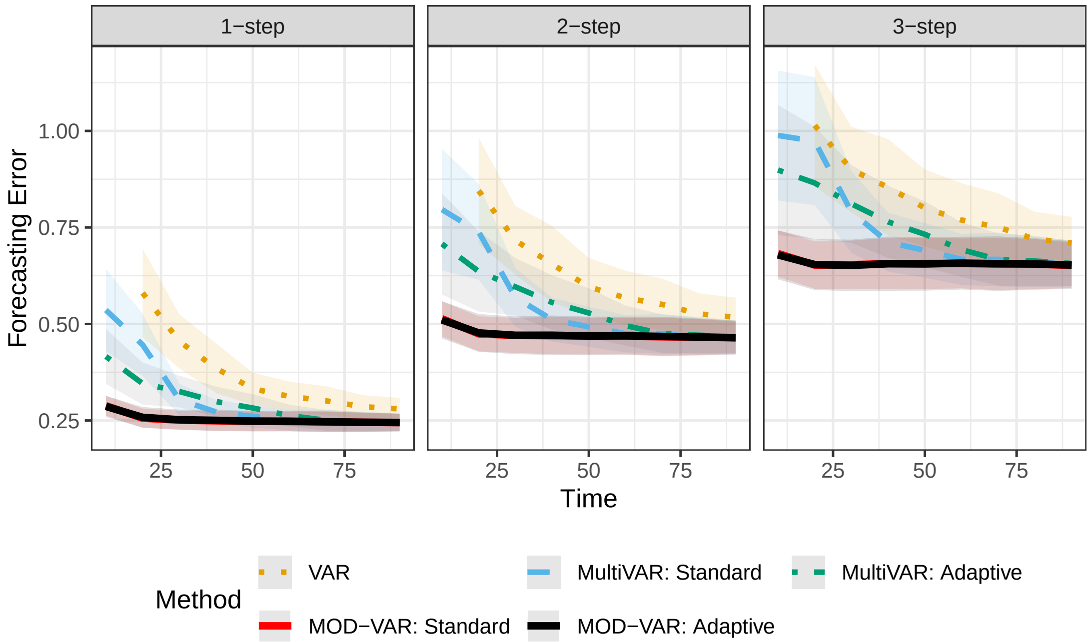


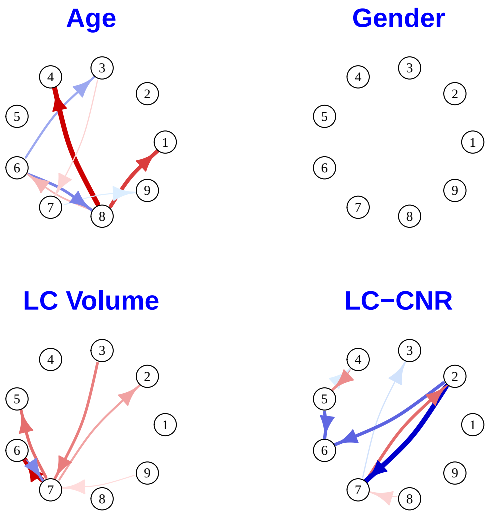
Figure 5: Comparison of the out-of-sample forecasting error of our proposed MOD-VAR, as well as the Multi-VAR and individual VAR models on the resting state fMRI dataset. By incorporating moderator information, our proposed MOD-VAR attains the lowest error for 1-, 2-, and 3-step forward forecasting performance.

coefficients $\{\hat{\Phi}^{(k)}\}_{k\in[n]}$. Then, we evaluate the out-of-sample k-th step forecasting error,

$$RMSFE_k(\widehat{\Phi}) = \left(\frac{1}{n}\sum_{k=1}^{n}\frac{1}{50}\sum_{h=1}^{50}\|\mathbf{Y}^{(k)}_{T_{train}+h-k} - \widehat{\Phi}^{(k)}\mathbf{Y}^{(k)}_{T_{train}+h}\|_2^2\right)^{1/2}.$$

To assess the quality of forecasting of the models, we consider the 1-, 2- and 3-step out-of-sample forecasting performance.

We visualize the resulting forecasting errors for the methods in Figure 5. Both the standard and adaptive MOD-VAR models exhibited the lowest forecasting error across 1-, 2-, and 3-step forward predictions thereby outperforming the standard VAR and Multi-VAR models. This result suggests that there is moderator-dependent structure along with participant-level heterogeneity in the resting state dynamics. We observe that both the MOD-VAR and Multi-VAR obtain a better performance than individually estimating VAR models, which suggest the presence of shared structure across subject. To estimate joint and individual structure, Multi-VAR must model a total of $d^2(1 + n) = 2187$ parameters.



Label	Summary	Region	Name
1	FPN_dlPFC	Frontoparietal Network	Dorsolateral prefrontal cortex
2	FPN_PPC		Posterior parietal cortex
3	FPN_dACC_pSMA		Dorsal anterior cingulate cortex/pre-supplementary motor area
4	DMN_mPFC	Default Mode Network	Medial prefrontal cortex
5	DMN_PCC_Pr		Posterior cingulate cortex/precuneus
6	DMN_IPL		Inferior parietal lobule
7	DMN_MTL		Medial temporal lobe
8	SN_AI	Salience Network	Anterior insula
9	SN_ACC		Anterior cingulate cortex

Figure 6: Visualization of the moderator effects estimated by our MOD-VAR method. Age moderates interactions of the salience network's anterior insula. Furthermore, locus coeruleous volume has network moderation effects, especially for the default mode network's medial temporal lobe. Locus coeruleus contrast-to-noise ratio moderates frotoparietal network's posterior pariental cortex. We found no network moderation effects for gender on the network dynamic.

On the other hand, since we have $p = 4$ moderators, our MOD-VAR using only moderators must estimate $d^2(1 + p) = 405$ parameters. Thus, by jointly modeling the multi-subject data, and reducing the complexity of the heterogeneity to be explained by the moderators, we attain visibly improved forecasting performance.

Next, we examined the estimated moderator effects obtained by our MOD-VAR estimates. We visualize the results in Figure6. Increasing age was associated with reduced predictive signaling between regions in the salience network (particularly anterior insula) and both the frontoparietal network (dorsolateral PFC) and the default mode network (medial prefrontal cortex). Further, predictive strength increased between the anterior insula and the inferior parietal lobule. Furthermore, we observed no moderation effects of gender on the predictive network dynamics.

Higher LC volume was associated with reduced predictive signaling within the default mode network regions, specifically from the medial temporal to both the posterior cingulate cortex/precuneus and the inferior parietal lobule. Reduced predictive strength was also observed from the frontoparietal network (dorsal anterior cingulate cortex/pre-supplementary motor area) to the default mode network (medial temporal lobe). Further, there was an increase in predictive strength from the inferior parietal lobule to the medial temporal lobe.

Higher LC contrast-to-noise ratio (CNR) was associated with increased predictive signaling from the frontoparietal network (posterior parietal cortex) to the default mode network (both the medial temporal lobe and the inferior parietal lobule). Increased predictive strength was also found from the posterior cingulate cortex/precuneus to the inferior parietal lobule. Conversely, elevated LC CNR was associated with reduced predictive signaling from the medial prefrontal cortex to the posterior cingulate cortex/precuneus, as well as from the medial temporal lobe to the posterior parietal cortex.

While the biological interpretations of these effects are outside the scope of the current paper, the results are somewhat consistent with theories of age-related declines in the LC and its associated role in moderating functional brain network activity. No gender effects were found which provides a type of discriminant validity for the MOD-VAR approach. On the one hand, prior work examining gender and resting state networks has found relationships while other work has suggested that these findings are highly dependent on data processing decisions. Finally, LC metrics were associated with predictive signaling in areas of the frontoparietal network and default mode network in a manner consistent with prior literature [24].

8 Discussion

In this work, we introduce the moderator vector auto-regressive model (MOD-VAR), for estimating the GCN associated to a set of heterogeneous multivariate time series data. Our proposed MOD-VAR can identify common structure across heterogeneous GNCs, and identify whether individual structure present in the model may be explained by external moderator variables. Our method outperforms methods in the literature in terms of both GNC recovery and forecasting error, as showcased by our extensive numerical experiments. Furthermore, our theoretical developments show that MOD-VAR attains fast rates of convergence, even in high-dimensional scenarios, as long as $n = \Omega(\log((n + p + 1)d^2))$.

This work opens a variety of new avenues of research. In particular, we consider important to further explore applications of our proposed MOD-VAR to other multi subject studies in psychology and neuroscience, similar to the works of [14; 25; 5]. Furthermore, our proposed method opens a variety of new avenues of methodological research. For example, it would be of interest to test the statistical significance of the effect of moderators on GCN networks. Further extensions beyond the sparsity assumption are also of interest.

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Supplementary Materials

We provide proofs, additional discussions and simulations in the Supplementary Materials.

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